# Manage network IP interfaces

ONTAP 9.10.1 REST API Documentation

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# Manage network IP interfaces

## Network IP interfaces endpoint overview

### Overview

The following operations are supported:

- Creation: POST network/ip/interfaces
- Collection Get: GET network/ip/interfaces
- Instance Get: GET network/ip/interfaces/{uuid}
- Instance Patch: PATCH network/ip/interfaces/{uuid}
- Instance Delete: DELETE network/ip/interfaces/{uuid}

### Retrieving network interface information

The IP interfaces GET API retrieves and displays relevant information pertaining to the interfaces configured in the cluster. The response can contain a list of multiple interfaces or a specific interface. The fields returned in the response vary for different interfaces and configurations.

### Examples

#### Retrieving all interfaces in the cluster

The following example shows the list of all interfaces configured in a cluster.

```
# The API:
/api/network/ip/interfaces

# The call:
curl -X GET "https://<mgmt-ip>/api/network/ip/interfaces" -H "accept:
application/hal+json"

# The response:
{
"records": [
  {
    "uuid": "14531286-59fc-11e8-ba55-005056b4340f",
    "name": "user-cluster-01_mgmt1",
    "_links": {
      "self": {
        "href": "/api/network/ip/interfaces/14531286-59fc-11e8-ba55-
005056b4340f"
      }
```

```
    }
  },
  {
    "uuid": "145318ba-59fc-11e8-ba55-005056b4340f",
    "name": "user-cluster-01_clus2",
    "_links": {
      "self": {
        "href": "/api/network/ip/interfaces/145318ba-59fc-11e8-ba55-
005056b4340f"
      }
    }
  },
  {
    "uuid": "14531e45-59fc-11e8-ba55-005056b4340f",
    "name": "user-cluster-01_clus1",
    "_links": {
      "self": {
        "href": "/api/network/ip/interfaces/14531e45-59fc-11e8-ba55-
005056b4340f"
      }
    }
  },
  {
    "uuid": "245979de-59fc-11e8-ba55-005056b4340f",
    "name": "cluster_mgmt",
    "_links": {
      "self": {
        "href": "/api/network/ip/interfaces/245979de-59fc-11e8-ba55-
005056b4340f"
      }
    }
  },
  {
    "uuid": "c670707c-5a11-11e8-8fcb-005056b4340f",
    "name": "lif1",
    "_links": {
      "self": {
        "href": "/api/network/ip/interfaces/c670707c-5a11-11e8-8fcb-
005056b4340f"
      }
    }
  }
],
"num_records": 5,
"_links": {
  "self": {
```

```
    "href": "/api/network/ip/interfaces"
  }
}
}
```

#### Retrieving a specific Cluster-scoped interface

The following example shows the response when a specific Cluster-scoped interface is requested. The system returns an error when there is no interface with the requested UUID. SVM information is not returned for Cluster-scoped interfaces.

```
# The API:
/api/network/ip/interfaces/{uuid}

# The call:
curl -X GET "https://<mgmt-ip>/api/network/ip/interfaces/245979de-59fc-
11e8-ba55-005056b4340f" -H "accept: application/hal+json"

# The response:
{
"uuid": "245979de-59fc-11e8-ba55-005056b4340f",
"name": "cluster_mgmt",
"ip": {
  "address": "10.63.41.6",
  "netmask": "18",
  "family": "ipv4",
},
"enabled": true,
"state": "up",
"scope": "cluster",
"ipspace": {
  "uuid": "114ecfb5-59fc-11e8-ba55-005056b4340f",
  "name": "Default",
  "_links": {
    "self": {
            "href": "/api/network/ipspaces/114ecfb5-59fc-11e8-ba55-
005056b4340f"
    }
  }
},
"services": [
  "management_core",
  "management_autosupport",
```

```
    "management_access"
  ],
  "location": {
    "is_home": true,
    "auto_revert": false,
    "failover": "broadcast_domain_only",
    "node": {
      "uuid": "c1db2904-1396-11e9-bb7d-005056acfcbb",
      "name": "user-cluster-01-a",
      "_links": {
        "self": {
          "href": "/api/cluster/nodes/c1db2904-1396-11e9-bb7d-005056acfcbb"
        }
      }
    },
    "port": {
      "uuid": "c84d5337-1397-11e9-87c2-005056acfcbb",
      "name": "e0d",
      "node": {
        "name": "user-cluster-01-a"
      },
      "_links": {
        "self": {
          "href": "/api/network/ethernet/ports/c84d5337-1397-11e9-87c2-
005056acfcbb"
        }
      }
    },
    "home_node": {
      "uuid": "c1db2904-1396-11e9-bb7d-005056acfcbb",
      "name": "user-cluster-01-a",
      "_links": {
        "self": {
          "href": "/api/cluster/nodes/c1db2904-1396-11e9-bb7d-005056acfcbb"
        }
      }
    },
    "home_port": {
      "uuid": "c84d5337-1397-11e9-87c2-005056acfcbb",
      "name": "e0d",
      "node": {
        "name": "user-cluster-01-a"
      },
      "_links": {
        "self": {
          "href": "/api/network/ethernet/ports/c84d5337-1397-11e9-87c2-
```

```
005056acfcbb"
      }
    }
  }
},
"service_policy": {
  "uuid": "9e0f4151-141b-11e9-851e-005056ac1ce0",
  "name": "default-management"
},
"vip": false,
"_links": {
  "self": {
    "href": "/api/network/ip/interfaces/245979de-59fc-11e8-ba55-
005056b4340f"
  }
}
}
```

#### Retrieving a specific SVM-scoped interface using a filter

The following example shows the response when a specific SVM-scoped interface is requested. The SVM object is only included for SVM-scoped interfaces.

```
# The API:
/api/network/ip/interfaces

# The call:
curl -X GET "https://<mgmt-
ip>/api/network/ip/interfaces?name=lif1&fields=*" -H "accept:
application/hal+json"

# The response:
{
"records": [
  {
    "uuid": "c670707c-5a11-11e8-8fcb-005056b4340f",
    "name": "lif1",
    "ip": {
      "address": "10.10.10.11",
      "netmask": "24",
      "family": "ipv4",
    },
    "enabled": true,
```

```
    "state": "up",
    "scope": "svm",
    "ipspace": {
      "uuid": "114ecfb5-59fc-11e8-ba55-005056b4340f",
      "name": "Default",
      "_links": {
        "self": {
          "href": "/api/network/ipspaces/114ecfb5-59fc-11e8-ba55-
005056b4340f"
        }
      }
    },
    "svm": {
      "uuid": "c2134665-5a11-11e8-8fcb-005056b4340f",
      "name": "user_vs0",
      "_links": {
        "self": {
          "href": "/api/svm/svms/c2134665-5a11-11e8-8fcb-005056b4340f"
        }
      }
    },
    "services": [
      "data_core",
      "data_nfs",
      "data_cifs",
      "data_flexcache"
    ],
    "location": {
      "is_home": true,
      "auto_revert": false,
      "failover": "broadcast_domain_only",
      "node": {
        "uuid": "c1db2904-1396-11e9-bb7d-005056acfcbb",
        "name": "user-cluster-01-a",
        "_links": {
          "self": {
            "href": "/api/cluster/nodes/c1db2904-1396-11e9-bb7d-
005056acfcbb"
          }
        }
      },
      "port": {
        "uuid": "c84d5337-1397-11e9-87c2-005056acfcbb",
        "name": "e0d",
        "node": {
          "name": "user-cluster-01-a"
```

```
        },
        "_links": {
          "self": {
            "href": "/api/network/ethernet/ports/c84d5337-1397-11e9-87c2-
005056acfcbb"
          }
        }
      },
      "home_node": {
        "uuid": "c1db2904-1396-11e9-bb7d-005056acfcbb",
        "name": "user-cluster-01-a",
        "_links": {
          "self": {
            "href": "/api/cluster/nodes/c1db2904-1396-11e9-bb7d-
005056acfcbb"
          }
        }
      },
      "home_port": {
        "uuid": "c84d5337-1397-11e9-87c2-005056acfcbb",
        "name": "e0d",
        "node": {
          "name": "user-cluster-01-a"
        },
        "_links": {
          "self": {
            "href": "/api/network/ethernet/ports/c84d5337-1397-11e9-87c2-
005056acfcbb"
          }
        }
      }
    },
    "service_policy": {
      "uuid": "9e53525f-141b-11e9-851e-005056ac1ce0",
      "name": "default-data-files"
    },
    "vip": false,
    "_links": {
      "self": {
        "href": "/api/network/ip/interfaces/c670707c-5a11-11e8-8fcb-
005056b4340f"
      }
    }
  }
],
"num_records": 1,
```

```
"_links": {
  "self": {
    "href": "/api/network/ip/interfaces?name=lif1&fields=*"
  }
}
}
```

#### Retrieving specific fields and limiting the output using filters

The following example shows the response when a filter is applied (location.home_port.name=e0a) and only certain fields are requested. Filtered fields are in the output in addition to the default fields and requested fields.

```
# The API:
/api/network/ip/interfaces

# The call:
curl -X GET "https://<mgmt-
ip>/api/network/ip/interfaces?location.home_port.name=e0a&fields=location.
home_node.name,service_policy.name,ip.address,enabled" -H "accept:
application/hal+json"

# The response:
{
"records": [
  {
    "uuid": "1d1c9dc8-4f17-11e9-9553-005056ac918a",
    "name": "user-cluster-01-a_clus1",
    "ip": {
      "address": "192.168.170.24"
    },
    "enabled": true,
    "location": {
      "home_node": {
        "name": "user-cluster-01-a"
      },
      "home_port": {
        "name": "e0a"
      }
    },
    "service_policy": {
      "name": "default-cluster"
    },
```

```
    "_links": {
      "self": {
        "href": "/api/network/ip/interfaces/1d1c9dc8-4f17-11e9-9553-
005056ac918a"
      }
    }
  },
  {
    "uuid": "d07782c1-4f16-11e9-86e7-005056ace7ee",
    "name": "user-cluster-01-b_clus1",
    "ip": {
      "address": "192.168.170.22"
    },
    "enabled": true,
    "location": {
      "home_node": {
        "name": "user-cluster-01-b"
      },
      "home_port": {
        "name": "e0a"
      }
    },
    "service_policy": {
      "name": "default-cluster"
    },
    "_links": {
      "self": {
        "href": "/api/network/ip/interfaces/d07782c1-4f16-11e9-86e7-
005056ace7ee"
      }
    }
  }
],
"num_records": 2,
"_links": {
  "self": {
    "href":
"/api/network/ip/interfaces?location.home_port.name=e0a&fields=location.ho
me_node.name,service_policy.name,ip.address,enabled"
  }
}
}
```

## Creating IP interfaces

You can use the IP interfaces POST API to create IP interfaces as shown in the following examples.

## Examples

### Creating a Cluster-scoped IP interface using names

The following example shows the record returned after the creation of an IP interface on "e0d".

```
# The API:
/api/network/ip/interfaces

# The call:
curl -X POST "https://<mgmt-
ip>/api/network/ip/interfaces?return_records=true" -H "accept:
application/hal+json" -d '{ "name": "cluster_mgmt", "ip": { "address":
"10.63.41.6", "netmask": "18" }, "enabled": true, "scope": "cluster",
"ipspace": { "name": "Default" }, "location": { "auto_revert": false,
"failover": "broadcast_domain_only", "home_port": { "name": "e0d", "node":
{ "name": "user-cluster-01-a" } } }, "service_policy": { "name": "default-
management" } }'

# The response:
{
"num_records": 1,
"records": [
  {
    "uuid": "245979de-59fc-11e8-ba55-005056b4340f",
    "name": "cluster_mgmt",
    "ip": {
      "address": "10.63.41.6",
      "netmask": "18"
    },
    "enabled": true,
    "scope": "cluster",
    "ipspace": {
      "name": "Default"
    },
    "location": {
      "auto_revert": false,
      "failover": "broadcast_domain_only",
      "home_port": {
        "name": "e0d",
```

```
        "node": {
          "name": "user-cluster-01-a"
        }
      },
    },
    "service_policy": {
      "name": "default-management"
    },
    "_links": {
      "self": {
        "href": "/api/network/ip/interfaces/245979de-59fc-11e8-ba55-
005056b4340f"
      }
    }
  }
]
}
```

#### Creating a SVM-scoped IP interface using a mix of parameter types

The following example shows the record returned after the creation of a IP interface by specifying a broadcast domain as the location.

```
# The API:
/api/network/ip/interfaces

# The call:
curl -X POST "https://<mgmt-
ip>/api/network/ip/interfaces?return_records=true" -H "accept:
application/hal+json" -d '{ "name": "Data1", "ip": { "address":
"10.234.101.116", "netmask": "255.255.240.0" }, "enabled": true, "scope":
"svm", "svm": { "uuid": "137f3618-1e89-11e9-803e-005056a7646a" },
"location": { "auto_revert": true, "broadcast_domain": { "name": "Default"
} }, "service_policy": { "name": "default-data-files" } }'

# The response:
{
"num_records": 1,
"records": [
  {
    "uuid": "80d271c9-1f43-11e9-803e-005056a7646a",
    "name": "Data1",
    "ip": {
```

```
      "address": "10.234.101.116",
      "netmask": "20"
    },
    "enabled": true,
    "scope": "svm",
    "svm": {
      "uuid": "137f3618-1e89-11e9-803e-005056a7646a",
      "name": "vs0",
      "_links": {
        "self": {
          "href": "/api/svm/svms/137f3618-1e89-11e9-803e-005056a7646a"
        }
      }
    },
    "location": {
      "auto_revert": true
    },
    "service_policy": {
      "name": "default-data-files"
    },
    "_links": {
      "self": {
        "href": "/api/network/ip/interfaces/80d271c9-1f43-11e9-803e-
005056a7646a"
      }
    }
  }
]
}
```


#### Creating a Cluster-scoped IP interface without specifying the scope parameter

The following example shows the record returned after creating an IP interface on "e0d" without specifying the scope parameter. The scope is "cluster" if an "svm" is not specified.

```
# The API:
/api/network/ip/interfaces

# The call:
curl -X POST "https://<mgmt-
ip>/api/network/ip/interfaces?return_records=true" -H "accept:
application/hal+json" -d '{ "name": "cluster_mgmt", "ip": { "address":
"10.63.41.6", "netmask": "18" }, "enabled": true, "ipspace": { "name":
```

```
"Default" }, "location": { "auto_revert": false, "home_port": { "name":
"e0d", "node": { "name": "user-cluster-01-a" } } }, "service_policy": {
"name": "default-management" } }'

# The response:
{
"num_records": 1,
"records": [
  {
    "uuid": "245979de-59fc-11e8-ba55-005056b4340f",
    "name": "cluster_mgmt",
    "ip": {
      "address": "10.63.41.6",
      "netmask": "18"
    },
    "enabled": true,
    "scope": "cluster",
    "ipspace": {
      "name": "Default"
    },
    "location": {
      "auto_revert": false,
      "home_port": {
        "name": "e0d",
        "node": {
          "name": "user-cluster-01-a"
        }
      }
    },
    "service_policy": {
      "name": "default-management"
    },
    "_links": {
      "self": {
        "href": "/api/network/ip/interfaces/245979de-59fc-11e8-ba55-
005056b4340f"
      }
    }
  }
]
}
```

#### Creating an SVM-scoped IP interface without specifying the scope parameter

The following example shows the record returned after creating an IP interface on "e0d" without specifying the scope parameter. The scope is "svm" if the "svm" field is specified.

```
# The API:
/api/network/ip/interfaces

# The call:
curl -X POST "https://<mgmt-
ip>/api/network/ip/interfaces?return_records=true" -H "accept:
application/hal+json" -d '{ "name": "Data1", "ip": { "address":
"10.234.101.116", "netmask": "255.255.240.0" }, "enabled": true, "svm": {
"uuid": "137f3618-1e89-11e9-803e-005056a7646a" }, "location": {
"auto_revert": true, "broadcast_domain": { "name": "Default" } },
"service_policy": { "name": "default-data-files" } }'

# The response:
{
"num_records": 1,
"records": [
  {
    "uuid": "80d271c9-1f43-11e9-803e-005056a7646a",
    "name": "Data1",
    "ip": {
      "address": "10.234.101.116",
      "netmask": "20"
    },
    "enabled": true,
    "scope": "svm",
    "svm": {
      "uuid": "137f3618-1e89-11e9-803e-005056a7646a",
      "name": "vs0",
      "_links": {
        "self": {
          "href": "/api/svms/137f3618-1e89-11e9-803e-005056a7646a"
        }
      }
    },
    "location": {
      "auto_revert": true
    },
    "service_policy": {
      "name": "default-data-files"
    },
```

```
    "_links": {
      "self": {
        "href": "/api/network/ip/interfaces/80d271c9-1f43-11e9-803e-
005056a7646a"
      }
    }
  }
]
}
```

## Updating IP interfaces

You can use the IP interfaces PATCH API to update the attributes of an IP interface.

## Examples

### Updating the auto revert flag of an IP interface

The following example shows how the PATCH request changes the auto revert flag to 'false'.

```
# The API:
/api/network/ip/interfaces/{uuid}

# The call:
curl -X PATCH "https://<mgmt-ip>/api/network/ip/interfaces/80d271c9-1f43-
11e9-803e-005056a7646a" -H "accept: application/hal+json" -d '{
"location": { "auto_revert": "false" } }'
{
}
```

### Updating the service policy of an IP interface

The following example shows how the PATCH request changes the service policy to 'default-management'.

```
# The API:
/api/network/ip/interfaces/{uuid}

# The call:
curl -X PATCH "https://<mgmt-ip>/api/network/ip/interfaces/80d271c9-1f43-
11e9-803e-005056a7646a" -H "accept: application/hal+json" -d '{
"service_policy": { "name": "default-management" }}'
{
}
```

## Deleting IP interfaces

You can use the IP interfaces DELETE API to delete an IP interface in the cluster.

## Example

### Deleting an IP Interface

The following DELETE request deletes a network IP interface.

```
# The API:
/api/network/ip/interfaces/{uuid}

# The call:
curl -X DELETE "https://<mgmt-ip>/api/network/ip/interfaces/80d271c9-1f43-
11e9-803e-005056a7646a"
{
}
```

# Retrieve all IP interface details

GET `/network/ip/interfaces`

**Introduced In:** 9.6

Retrieves the details of all IP interfaces.

## Related ONTAP Commands

- `network interface show`

## Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| ipspace.uuid | string | query | False | Filter by ipspace.uuid |
| ipspace.name | string | query | False | Filter by ipspace.name |
| services | string | query | False | Filter by services |
| svm.uuid | string | query | False | Filter by svm.uuid |
| svm.name | string | query | False | Filter by svm.name |
| ddns_enabled | boolean | query | False | Filter by ddns_enabled<br>• Introduced in: 9.9 |
| scope | string | query | False | Filter by scope |
| vip | boolean | query | False | Filter by vip |
| state | string | query | False | Filter by state |
| ip.address | string | query | False | Filter by ip.address |
| ip.family | string | query | False | Filter by ip.family |
| ip.netmask | string | query | False | Filter by ip.netmask |
| dns_zone | string | query | False | Filter by dns_zone<br>• Introduced in: 9.9 |
| location.failover | string | query | False | Filter by location.failover |

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| location.home_node.uuid | string | query | False | Filter by location.home_node.uuid |
| location.home_node.name | string | query | False | Filter by location.home_node.name |
| location.node.uuid | string | query | False | Filter by location.node.uuid |
| location.node.name | string | query | False | Filter by location.node.name |
| location.home_port.name | string | query | False | Filter by location.home_port.name |
| location.home_port.node.name | string | query | False | Filter by location.home_port.node.name |
| location.home_port.uuid | string | query | False | Filter by location.home_port.uuid |
| location.port.name | string | query | False | Filter by location.port.name |
| location.port.node.name | string | query | False | Filter by location.port.node.name |
| location.port.uuid | string | query | False | Filter by location.port.uuid |
| location.auto_revert | boolean | query | False | Filter by location.auto_revert |
| location.is_home | boolean | query | False | Filter by location.is_home |
| statistics.status | string | query | False | Filter by statistics.status<br>• Introduced in: 9.8 |

| Name | Type | In | Required | Description |
| --- | --- | --- | --- | --- |
| statistics.timestamp | string | query | False | Filter by statistics.timestamp<br>• Introduced in: 9.8 |
| statistics.throughput_raw.read | integer | query | False | Filter by statistics.throughput_raw.read<br>• Introduced in: 9.8 |
| statistics.throughput_raw.total | integer | query | False | Filter by statistics.throughput_raw.total<br>• Introduced in: 9.8 |
| statistics.throughput_raw.write | integer | query | False | Filter by statistics.throughput_raw.write<br>• Introduced in: 9.8 |
| service_policy.name | string | query | False | Filter by service_policy.name |
| service_policy.uuid | string | query | False | Filter by service_policy.uuid |
| enabled | boolean | query | False | Filter by enabled |
| metric.duration | string | query | False | Filter by metric.duration<br>• Introduced in: 9.8 |
| metric.timestamp | string | query | False | Filter by metric.timestamp<br>• Introduced in: 9.8 |

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| metric.status | string | query | False | Filter by metric.status<br>• Introduced in: 9.8 |
| metric.throughput.read | integer | query | False | Filter by metric.throughput.read<br>• Introduced in: 9.8 |
| metric.throughput.total | integer | query | False | Filter by metric.throughput.total<br>• Introduced in: 9.8 |
| metric.throughput.write | integer | query | False | Filter by metric.throughput.write<br>• Introduced in: 9.8 |
| probe_port | integer | query | False | Filter by probe_port<br>• Introduced in: 9.10 |
| rdma_protocols | string | query | False | Filter by rdma_protocols<br>• Introduced in: 9.10 |
| name | string | query | False | Filter by name |
| uuid | string | query | False | Filter by uuid |
| fields | array[string] | query | False | Specify the fields to return. |
| max_records | integer | query | False | Limit the number of records returned. |

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| return_records | boolean | query | False | The default is true for GET calls. When set to false, only the number of records is returned.<br>• Default value: 1 |
| return_timeout | integer | query | False | The number of seconds to allow the call to execute before returning. When iterating over a collection, the default is 15 seconds. ONTAP returns earlier if either max records or the end of the collection is reached.<br>• Default value: 1<br>• Max value: 120<br>• Min value: 0 |
| order_by | array[string] | query | False | Order results by specified fields and optional [asc |

## Response

```
Status: 200, Ok
```

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| num_records | integer | |
| records | array[ip_interface] | |

**Example response**


```
{
  "_links": {
    "next": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "num_records": 1,
  "records": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "dns_zone": "storage.company.com",
    "ip": {
      "address": "10.10.10.7",
      "family": "ipv4",
      "netmask": "24"
    },
    "ipspace": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "exchange",
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "location": {
      "broadcast_domain": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "name": "bd1",
        "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
      },
      "failover": "home_port_only",
      "home_node": {
        "_links": {
```

```
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "name": "node1",
        "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
      },
      "home_port": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "name": "e1b",
        "node": {
          "name": "node1"
        },
        "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
      },
      "node": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "name": "node1",
        "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
      },
      "port": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "name": "e1b",
        "node": {
          "name": "node1"
        },
        "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
      }
    },
    "metric": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
```

```
      },
      "duration": "PT15S",
      "status": "ok",
      "throughput": {
        "read": 200,
        "total": 1000,
        "write": 100
      },
      "timestamp": "2017-01-25T11:20:13Z"
    },
    "name": "dataLif1",
    "probe_port": 64001,
    "rdma_protocols": {
    },
    "scope": "svm",
    "service_policy": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "default-intercluster",
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "services": {
    },
    "state": "up",
    "statistics": {
      "status": "ok",
      "throughput_raw": {
        "read": 200,
        "total": 1000,
        "write": 100
      },
      "timestamp": "2017-01-25T11:20:13Z"
    },
    "svm": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "svm1",
      "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
    },
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
```

```
    }
  }
```

## Error

```
Status: Default, Error
```

| Name | Type | Description |
|---|---|---|
| error | error | |

### Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

## Definitions

**See Definitions**

href

| Name | Type | Description |
|---|---|---|
| href | string | |

_links

| Name | Type | Description |
|---|---|---|
| next | href | |
| self | href | |

_links

| Name | Type | Description |
|---|---|---|
| self | href | |

ip_info

IP information

| Name | Type | Description |
|---|---|---|
| address | string | IPv4 or IPv6 address |
| family | string | IPv4 or IPv6 |
| netmask | string | Input as netmask length (16) or IPv4 mask (255.255.0.0). For IPv6, the default value is 64 with a valid range of 1 to 127. Output is always netmask length. |

ipspace

Either the UUID or name must be supplied on POST for cluster-scoped objects.

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | IPspace name |
| uuid | string | IPspace UUID |

broadcast_domain

Broadcast domain UUID along with a readable name.

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | Name of the broadcast domain, scoped to its IPspace |
| uuid | string | Broadcast domain UUID |

home_node

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | |
| uuid | string | |

node

| Name | Type | Description |
|---|---|---|
| name | string | Name of node on which the port is located. |

home_port

Port UUID along with readable names. Either the UUID or both names may be supplied on input.

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | |
| node | node | |
| uuid | string | |

node

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | |
| uuid | string | |

port

Port UUID along with readable names. Either the UUID or both names may be supplied on input.

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | |
| node | node | |
| uuid | string | |

location

Current or home location can be modified. Specifying a port implies a node. Specifying a node allows an appropriate port to be automatically selected. Ports are not valid and not shown for VIP interfaces. For POST, broadcast_domain can be specified alone or with home_node. For PATCH, set is_home to true to revert a LIF back to its home port.

| Name | Type | Description |
|---|---|---|
| auto_revert | boolean | |
| broadcast_domain | broadcast_domain | Broadcast domain UUID along with a readable name. |
| failover | string | Defines where an interface may failover. |
| home_node | home_node | |
| home_port | home_port | Port UUID along with readable names. Either the UUID or both names may be supplied on input. |
| is_home | boolean | |
| node | node | |
| port | port | Port UUID along with readable names. Either the UUID or both names may be supplied on input. |

throughput

The rate of throughput bytes per second observed at the interface.

| Name | Type | Description |
|---|---|---|
| read | integer | Performance metric for read I/O operations. |
| total | integer | Performance metric aggregated over all types of I/O operations. |

| Name | Type | Description |
| --- | --- | --- |
| write | integer | Peformance metric for write I/O operations. |

metric

The most recent sample of I/O metrics for the interface.

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| duration | string | The duration over which this sample is calculated. The time durations are represented in the ISO-8601 standard format. Samples can be calculated over the following durations: |
| status | string | Errors associated with the sample. For example, if the aggregation of data over multiple nodes fails, then any partial errors might return "ok" on success or "error" on an internal uncategorized failure. Whenever a sample collection is missed but done at a later time, it is back filled to the previous 15 second timestamp and tagged with "backfilled_data". "inconsistent_delta_time" is encountered when the time between two collections is not the same for all nodes. Therefore, the aggregated value might be over or under inflated. "Negative_delta" is returned when an expected monotonically increasing value has decreased in value. "inconsistent_old_data" is returned when one or more nodes do not have the latest data. |
| throughput | throughput | The rate of throughput bytes per second observed at the interface. |
| timestamp | string | The timestamp of the performance data. |

service_policy

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | |
| uuid | string | |

throughput_raw

Throughput bytes observed at the interface. This can be used along with delta time to calculate the rate of throughput bytes per unit of time.

| Name | Type | Description |
|---|---|---|
| read | integer | Performance metric for read I/O operations. |
| total | integer | Performance metric aggregated over all types of I/O operations. |
| write | integer | Peformance metric for write I/O operations. |

statistics

The real time I/O statistics for the interface.

| Name | Type | Description |
| --- | --- | --- |
| status | string | Errors associated with the sample. For example, if the aggregation of data over multiple nodes fails, then any partial errors might return "ok" on success or "error" on an internal uncategorized failure. Whenever a sample collection is missed but done at a later time, it is back filled to the previous 15 second timestamp and tagged with "backfilled_data". "inconsistent_delta_time" is encountered when the time between two collections is not the same for all nodes. Therefore, the aggregated value might be over or under inflated. "Negative_delta" is returned when an expected monotonically increasing value has decreased in value. "inconsistent_old_data" is returned when one or more nodes do not have the latest data. |
| throughput_raw | throughput_raw | Throughput bytes observed at the interface. This can be used along with delta time to calculate the rate of throughput bytes per unit of time. |
| timestamp | string | The timestamp of the performance data. |

svm

Applies only to SVM-scoped objects. Either the UUID or name must be supplied on POST.

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| name | string | The name of the SVM. |
| uuid | string | The unique identifier of the SVM. |

ip_interface

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| ddns_enabled | boolean | Indicates whether or not dynamic DNS updates are enabled. Defaults to true if the interface supports "data_nfs" or "data_cifs" services, otherwise false. |
| dns_zone | string | Fully qualified DNS zone name |
| enabled | boolean | The administrative state of the interface. |
| ip | ip_info | IP information |
| ipspace | ipspace | Either the UUID or name must be supplied on POST for cluster-scoped objects. |
| location | location | Current or home location can be modified. Specifying a port implies a node. Specifying a node allows an appropriate port to be automatically selected. Ports are not valid and not shown for VIP interfaces. For POST, broadcast_domain can be specified alone or with home_node. For PATCH, set is_home to true to revert a LIF back to its home port. |
| metric | metric | The most recent sample of I/O metrics for the interface. |
| name | string | Interface name |
| probe_port | integer | Probe port for Cloud load balancer |
| rdma_protocols | array[string] | Supported RDMA offload protocols |
| scope | string | Set to "svm" for interfaces owned by an SVM. Otherwise, set to "cluster". |
| service_policy | service_policy | |

| Name | Type | Description |
|---|---|---|
| services | array[string] | The services associated with the interface. |
| state | string | The operational state of the interface. |
| statistics | statistics | The real time I/O statistics for the interface. |
| svm | svm | Applies only to SVM-scoped objects. Either the UUID or name must be supplied on POST. |
| uuid | string | The UUID that uniquely identifies the interface. |
| vip | boolean | True for a VIP interface, whose location is announced via BGP. |

error_arguments

| Name | Type | Description |
|---|---|---|
| code | string | Argument code |
| message | string | Message argument |

error

| Name | Type | Description |
|---|---|---|
| arguments | array[error_arguments] | Message arguments |
| code | string | Error code |
| message | string | Error message |
| target | string | The target parameter that caused the error. |

# Create a new cluster-scoped or SVM-scoped interface

POST `/network/ip/interfaces`

**Introduced In:** 9.6

Creates a new Cluster-scoped or SVM-scoped interface.

## Required properties

- `name` - Name of the interface to create.
- `ip.address` - IP address for the interface.
- `ip.netmask` - IP subnet of the interface.
- `ipspace.name` or `ipspace.uuid`
- Required for Cluster-scoped interfaces.
- Optional for SVM-scoped interfaces.
- `svm.name` or `svm.uuid`
- Required for an SVM-scoped interface.
- Invalid for a Cluster-scoped interface.
- `location.home_port` or `location.home_node` or `location.broadcast_domain` - One of these properties must be set to a value to define where the interface will be located.

## Recommended property values

- `service_policy`
- `for SVM scoped interfaces`
- *default-data-files* for interfaces carrying file-oriented NAS data traffic
- *default-data-blocks* for interfaces carrying block-oriented SAN data traffic
- *default-management* for interfaces carrying SVM management requests
- `for Cluster scoped interfaces`
- *default-intercluster* for interfaces carrying cluster peering traffic
- *default-management* for interfaces carrying system management requests
- *default-route-announce* for interfaces carrying BGP peer connections

## Default property values

If not specified in POST, the following default property values are assigned:

- `scope`
- *svm* if svm parameter is specified.
- *cluster* if svm parameter is not specified
- `enabled` - *true*
- `location.auto_revert` - *true*
- `service_policy`
- *default-data-files* if scope is `svm`

- *default-management* if scope is `cluster` and IPspace is not `Cluster`
- *default-cluster* if scope is `svm` and IPspace is `Cluster`
- `failover` - Selects the least restrictive failover policy supported by all the services in the service policy.
- `ddns_enabled`
- *true* if the interface supports *data_nfs* or *data_cifs* services
- *false* otherwise

## Related ONTAP commands

- `network interface create`

## Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| return_records | boolean | query | False | The default is false. If set to true, the records are returned.<br>• Default value: |

## Request Body

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| ddns_enabled | boolean | Indicates whether or not dynamic DNS updates are enabled. Defaults to true if the interface supports "data_nfs" or "data_cifs" services, otherwise false. |
| dns_zone | string | Fully qualified DNS zone name |
| enabled | boolean | The administrative state of the interface. |
| ip | ip_info | IP information |
| ipspace | ipspace | Either the UUID or name must be supplied on POST for cluster-scoped objects. |

| Name | Type | Description |
| --- | --- | --- |
| location | location | Current or home location can be modified. Specifying a port implies a node. Specifying a node allows an appropriate port to be automatically selected. Ports are not valid and not shown for VIP interfaces. For POST, broadcast_domain can be specified alone or with home_node. For PATCH, set is_home to true to revert a LIF back to its home port. |
| metric | metric | The most recent sample of I/O metrics for the interface. |
| name | string | Interface name |
| probe_port | integer | Probe port for Cloud load balancer |
| rdma_protocols | array[string] | Supported RDMA offload protocols |
| scope | string | Set to "svm" for interfaces owned by an SVM. Otherwise, set to "cluster". |
| service_policy | service_policy | |
| services | array[string] | The services associated with the interface. |
| state | string | The operational state of the interface. |
| statistics | statistics | The real time I/O statistics for the interface. |
| svm | svm | Applies only to SVM-scoped objects. Either the UUID or name must be supplied on POST. |
| uuid | string | The UUID that uniquely identifies the interface. |
| vip | boolean | True for a VIP interface, whose location is announced via BGP. |

**Example request**


```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "dns_zone": "storage.company.com",
  "ip": {
    "address": "10.10.10.7",
    "family": "ipv4",
    "netmask": "24"
  },
  "ipspace": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "exchange",
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
  },
  "location": {
    "broadcast_domain": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "bd1",
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "failover": "home_port_only",
    "home_node": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "node1",
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "home_port": {
      "_links": {
        "self": {
```

```
          "href": "/api/resourcelink"
        }
      },
      "name": "e1b",
      "node": {
        "name": "node1"
      },
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "node": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "node1",
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "port": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "e1b",
      "node": {
        "name": "node1"
      },
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    }
  },
  "metric": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "duration": "PT15S",
    "status": "ok",
    "throughput": {
      "read": 200,
      "total": 1000,
      "write": 100
    },
    "timestamp": "2017-01-25T11:20:13Z"
  },
```

```
  "name": "dataLif1",
  "probe_port": 64001,
  "rdma_protocols": {
  },
  "scope": "svm",
  "service_policy": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "default-intercluster",
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
  },
  "services": {
  },
  "state": "up",
  "statistics": {
    "status": "ok",
    "throughput_raw": {
      "read": 200,
      "total": 1000,
      "write": 100
    },
    "timestamp": "2017-01-25T11:20:13Z"
  },
  "svm": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "svm1",
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  },
  "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
}
```


## Response

```
Status: 201, Created
```

## Error

```
Status: Default
```

ONTAP Error Response Codes

| Error Code | Description |
|---|---|
| 1376656 | Cluster interfaces must be in the same subnet. Verify the address and netmask are set to the correct values. |
| 1376663 | All LIFs from a single DNS zone must be in the same SVM. |
| 1376663 | Cannot add interface to DNS zone because all interfaces from a single DNS zone must be in the same SVM. |
| 1376963 | Duplicate IP address. |
| 1966138 | The same IP address may not be used for both a mgmt interface and a gateway address. |
| 1966140 | An interface with the same name already exists. |
| 1966141 | Invalid DNS zone name. |
| 1966142 | Only data LIFs can be assigned a DNS zone. |
| 1966267 | IPv6 addresses must have a prefix length between 1 and 127. |
| 1966269 | IPv4 addresses must have a prefix length between 1 and 32. |
| 1966270 | Operation not support on SAN LIFs. |
| 1966476 | DNS Update is supported only on data LIFs. |
| 1966477 | DNS Update is supported only on LIFs configured with the NFS or CIFS protocol. |
| 1966987 | The Vserver Broadcast-Domain Home-Node and Home-Port combination is not valid. |
| 1967081 | The specified SVM must exist in the specified IPspace. |
| 1967082 | The specified ipspace.name does not match the IPspace name of ipspace.uuid. |
| 1967102 | POST operation might have left configuration in an inconsistent state. Check the configuration. |
| 1967106 | The specified location.home_port.name does not match the specified port name of location.home_port.uuid. |
| 1967107 | The location.home_port.uuid specified is not valid. |

| Error Code | Description |
| --- | --- |
| 1967108 | The specified location.home_node.name does not match the node name of location.home_node.uuid. |
| 1967109 | The specified location.home_port.node.name does not match the node name of location.home_node.uuid. |
| 1967110 | The specified location.home_port.node.name does not match location.home_node.name. |
| 1967111 | Home node must be specified by at least one location.home_node, location.home_port, or location.broadcast_domain field. |
| 1967112 | The specified location.home_node.name does not match the node name of location.home_port.uuid. |
| 1967120 | The specified service_policy.name does not match the specified service policy name of service_policy.uuid. |
| 1967121 | Invalid service_policy.uuid specified. |
| 1967122 | The specified location.broadcast_domain.name does not match the specified broadcast domain name of location.broadcast_domain.uuid. |
| 1967123 | The specified IPspace does not match the IPspace name of location.broadcast_domain.uuid. |
| 1967124 | The location.broadcast_domain.uuid specified is not valid. |
| 1967127 | svm.uuid or svm.name must be provided if scope is "svm". |
| 1967128 | ipspace.uuid or ipspace.name must be provided if scope is "cluster". |
| 1967129 | The specified location.home_port.uuid is not valid. |
| 1967130 | The specified location.home_port.name is not valid. |
| 1967131 | The specified location.home_port.uuid and location.home_port.name are not valid. |
| 1967135 | The specified location.broadcast_domain.uuid is not valid. |
| 1967136 | The specified location.broadcast_domain.name (and ipspace name) is not valid. |
| 1967137 | The specified location.broadcast_domain.uuid and location.broadcast_domain.name (and IPspace name) are not valid. |
| 1967145 | The specified location.failover is not valid. |
| 1967146 | The specified svm.name is not valid. |
| 1967147 | The specified svm.uuid is not valid. |

| Error Code | Description |
| --- | --- |
| 1967153 | No suitable port exists on location.home_node to host the interface. |
| 1967154 | Interfaces cannot be created on ports that are down. If a broadcast domain is specified, ensure that it contains at least one port that is operationally up. |
| 1967381 | Post VIP interfaces requires an effective cluster version of 9.7 or later. |
| 1967382 | VIP interfaces only reside in SVM scope. |
| 1967383 | Neither location.home_port.uuid or location.home_port.name should be set with vip=true. |
| 1967384 | Failed to create VIP interface because the home node does not have active BGP sessions to support Virtual IP (VIP) traffic. |
| 1967385 | VIP interfaces with an IPv4 address must use ip.netmask=32. VIP interfaces with an IPv6 address must use ip.netmask=128. |
| 1967387 | The specified IP address is in use by a subnet in this IPspace. |
| 1967391 | Setting the DNS zone requires an effective cluster version of 9.9.1 or later. |
| 1967392 | Setting the DDNS enable parameter requires an effective cluster version of 9.9.1 or later. |
| 1967394 | Setting the probe port parameter requires an effective cluster version of 9.10.1 or later. |
| 5373966 | An iSCSI interface cannot be created in an SVM configured for NVMe. |
| 53281036 | Setting the probe port parameter is not allowed on this platform. |
| 53281065 | The service_policy does not exist in the SVM. |
| 53281086 | LIF would exceed the maximum number of supported intercluster LIFs in IPspace. |
| 53281087 | Cannot configure SAN LIF on SVM. |

| Name | Type | Description |
| --- | --- | --- |
| error | error | |

**Example error**

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

## Definitions

**See Definitions**

href

| Name | Type | Description |
|---|---|---|
| href | string | |

_links

| Name | Type | Description |
|---|---|---|
| self | href | |

ip_info

IP information

| Name | Type | Description |
|---|---|---|
| address | string | IPv4 or IPv6 address |
| family | string | IPv4 or IPv6 |
| netmask | string | Input as netmask length (16) or IPv4 mask (255.255.0.0). For IPv6, the default value is 64 with a valid range of 1 to 127. Output is always netmask length. |

ipspace

Either the UUID or name must be supplied on POST for cluster-scoped objects.

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | IPspace name |
| uuid | string | IPspace UUID |

broadcast_domain

Broadcast domain UUID along with a readable name.

| Name | Type | Description |
|---|---|---|
| _links | _links | |

| Name | Type | Description |
| --- | --- | --- |
| name | string | Name of the broadcast domain, scoped to its IPspace |
| uuid | string | Broadcast domain UUID |

home_node

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| name | string | |
| uuid | string | |

node

| Name | Type | Description |
| --- | --- | --- |
| name | string | Name of node on which the port is located. |

home_port

Port UUID along with readable names. Either the UUID or both names may be supplied on input.

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| name | string | |
| node | node | |
| uuid | string | |

node

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| name | string | |
| uuid | string | |

port

Port UUID along with readable names. Either the UUID or both names may be supplied on input.

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |

| Name | Type | Description |
|---|---|---|
| name | string | |
| node | node | |
| uuid | string | |

location

Current or home location can be modified. Specifying a port implies a node. Specifying a node allows an appropriate port to be automatically selected. Ports are not valid and not shown for VIP interfaces. For POST, broadcast_domain can be specified alone or with home_node. For PATCH, set is_home to true to revert a LIF back to its home port.

| Name | Type | Description |
|---|---|---|
| auto_revert | boolean | |
| broadcast_domain | broadcast_domain | Broadcast domain UUID along with a readable name. |
| failover | string | Defines where an interface may failover. |
| home_node | home_node | |
| home_port | home_port | Port UUID along with readable names. Either the UUID or both names may be supplied on input. |
| is_home | boolean | |
| node | node | |
| port | port | Port UUID along with readable names. Either the UUID or both names may be supplied on input. |

throughput

The rate of throughput bytes per second observed at the interface.

| Name | Type | Description |
|---|---|---|
| read | integer | Performance metric for read I/O operations. |
| total | integer | Performance metric aggregated over all types of I/O operations. |
| write | integer | Peformance metric for write I/O operations. |

metric

The most recent sample of I/O metrics for the interface.

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| duration | string | The duration over which this sample is calculated. The time durations are represented in the ISO-8601 standard format. Samples can be calculated over the following durations: |
| status | string | Errors associated with the sample. For example, if the aggregation of data over multiple nodes fails, then any partial errors might return "ok" on success or "error" on an internal uncategorized failure. Whenever a sample collection is missed but done at a later time, it is back filled to the previous 15 second timestamp and tagged with "backfilled_data". "inconsistent_delta_time" is encountered when the time between two collections is not the same for all nodes. Therefore, the aggregated value might be over or under inflated. "Negative_delta" is returned when an expected monotonically increasing value has decreased in value. "inconsistent_old_data" is returned when one or more nodes do not have the latest data. |
| throughput | throughput | The rate of throughput bytes per second observed at the interface. |
| timestamp | string | The timestamp of the performance data. |

service_policy

| Name | Type | Description |
|---|---|---|
| _links | _links | |

| Name | Type | Description |
| --- | --- | --- |
| name | string | |
| uuid | string | |

throughput_raw

Throughput bytes observed at the interface. This can be used along with delta time to calculate the rate of throughput bytes per unit of time.

| Name | Type | Description |
| --- | --- | --- |
| read | integer | Performance metric for read I/O operations. |
| total | integer | Performance metric aggregated over all types of I/O operations. |
| write | integer | Peformance metric for write I/O operations. |

statistics

The real time I/O statistics for the interface.

| Name | Type | Description |
| --- | --- | --- |
| status | string | Errors associated with the sample. For example, if the aggregation of data over multiple nodes fails, then any partial errors might return "ok" on success or "error" on an internal uncategorized failure. Whenever a sample collection is missed but done at a later time, it is back filled to the previous 15 second timestamp and tagged with "backfilled_data". "inconsistent_delta_time" is encountered when the time between two collections is not the same for all nodes. Therefore, the aggregated value might be over or under inflated. "Negative_delta" is returned when an expected monotonically increasing value has decreased in value. "inconsistent_old_data" is returned when one or more nodes do not have the latest data. |
| throughput_raw | throughput_raw | Throughput bytes observed at the interface. This can be used along with delta time to calculate the rate of throughput bytes per unit of time. |
| timestamp | string | The timestamp of the performance data. |

svm

Applies only to SVM-scoped objects. Either the UUID or name must be supplied on POST.

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| name | string | The name of the SVM. |
| uuid | string | The unique identifier of the SVM. |

ip_interface

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| ddns_enabled | boolean | Indicates whether or not dynamic DNS updates are enabled. Defaults to true if the interface supports "data_nfs" or "data_cifs" services, otherwise false. |
| dns_zone | string | Fully qualified DNS zone name |
| enabled | boolean | The administrative state of the interface. |
| ip | ip_info | IP information |
| ipspace | ipspace | Either the UUID or name must be supplied on POST for cluster-scoped objects. |
| location | location | Current or home location can be modified. Specifying a port implies a node. Specifying a node allows an appropriate port to be automatically selected. Ports are not valid and not shown for VIP interfaces. For POST, broadcast_domain can be specified alone or with home_node. For PATCH, set is_home to true to revert a LIF back to its home port. |
| metric | metric | The most recent sample of I/O metrics for the interface. |
| name | string | Interface name |
| probe_port | integer | Probe port for Cloud load balancer |
| rdma_protocols | array[string] | Supported RDMA offload protocols |
| scope | string | Set to "svm" for interfaces owned by an SVM. Otherwise, set to "cluster". |
| service_policy | service_policy | |

| Name | Type | Description |
|---|---|---|
| services | array[string] | The services associated with the interface. |
| state | string | The operational state of the interface. |
| statistics | statistics | The real time I/O statistics for the interface. |
| svm | svm | Applies only to SVM-scoped objects. Either the UUID or name must be supplied on POST. |
| uuid | string | The UUID that uniquely identifies the interface. |
| vip | boolean | True for a VIP interface, whose location is announced via BGP. |

error_arguments

| Name | Type | Description |
|---|---|---|
| code | string | Argument code |
| message | string | Message argument |

error

| Name | Type | Description |
|---|---|---|
| arguments | array[error_arguments] | Message arguments |
| code | string | Error code |
| message | string | Error message |
| target | string | The target parameter that caused the error. |

# Delete an IP interface

DELETE `/network/ip/interfaces/{uuid}`

**Introduced In:** 9.6

Deletes an IP interface.

### Related ONTAP commands

- `network interface delete`

### Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| uuid | string | path | True | IP interface UUID |

### Response

```
Status: 200, Ok
```

## Retrieve details for an IP interface

GET `/network/ip/interfaces/{uuid}`

**Introduced In:** 9.6

Retrieves details for a specific IP interface.

### Related ONTAP commands

- `network interface show`

### Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| uuid | string | path | True | IP interface UUID |
| fields | array[string] | query | False | Specify the fields to return. |

### Response

```
Status: 200, Ok
```

| Name | Type | Description |
|---|---|---|
| _links | _links | |

| Name | Type | Description |
| --- | --- | --- |
| ddns_enabled | boolean | Indicates whether or not dynamic DNS updates are enabled. Defaults to true if the interface supports "data_nfs" or "data_cifs" services, otherwise false. |
| dns_zone | string | Fully qualified DNS zone name |
| enabled | boolean | The administrative state of the interface. |
| ip | ip_info | IP information |
| ipspace | ipspace | Either the UUID or name must be supplied on POST for cluster-scoped objects. |
| location | location | Current or home location can be modified. Specifying a port implies a node. Specifying a node allows an appropriate port to be automatically selected. Ports are not valid and not shown for VIP interfaces. For POST, broadcast_domain can be specified alone or with home_node. For PATCH, set is_home to true to revert a LIF back to its home port. |
| metric | metric | The most recent sample of I/O metrics for the interface. |
| name | string | Interface name |
| probe_port | integer | Probe port for Cloud load balancer |
| rdma_protocols | array[string] | Supported RDMA offload protocols |
| scope | string | Set to "svm" for interfaces owned by an SVM. Otherwise, set to "cluster". |
| service_policy | service_policy | |
| services | array[string] | The services associated with the interface. |

| Name | Type | Description |
| --- | --- | --- |
| state | string | The operational state of the interface. |
| statistics | statistics | The real time I/O statistics for the interface. |
| svm | svm | Applies only to SVM-scoped objects. Either the UUID or name must be supplied on POST. |
| uuid | string | The UUID that uniquely identifies the interface. |
| vip | boolean | True for a VIP interface, whose location is announced via BGP. |

**Example response**


```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "dns_zone": "storage.company.com",
  "ip": {
    "address": "10.10.10.7",
    "family": "ipv4",
    "netmask": "24"
  },
  "ipspace": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "exchange",
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
  },
  "location": {
    "broadcast_domain": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "bd1",
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "failover": "home_port_only",
    "home_node": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "node1",
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "home_port": {
      "_links": {
        "self": {
```

```
          "href": "/api/resourcelink"
        }
      },
      "name": "e1b",
      "node": {
        "name": "node1"
      },
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "node": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "node1",
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "port": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "e1b",
      "node": {
        "name": "node1"
      },
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    }
  },
  "metric": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "duration": "PT15S",
    "status": "ok",
    "throughput": {
      "read": 200,
      "total": 1000,
      "write": 100
    },
    "timestamp": "2017-01-25T11:20:13Z"
  },
```

```
  "name": "dataLif1",
  "probe_port": 64001,
  "rdma_protocols": {
  },
  "scope": "svm",
  "service_policy": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "default-intercluster",
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
  },
  "services": {
  },
  "state": "up",
  "statistics": {
    "status": "ok",
    "throughput_raw": {
      "read": 200,
      "total": 1000,
      "write": 100
    },
    "timestamp": "2017-01-25T11:20:13Z"
  },
  "svm": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "svm1",
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  },
  "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
}
```


## Error

```
Status: Default, Error
```

| Name | Type | Description |
| --- | --- | --- |
| error | error | |

**Example error**

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

## Definitions

**See Definitions**

href

| Name | Type | Description |
|---|---|---|
| href | string | |

_links

| Name | Type | Description |
|---|---|---|
| self | href | |

ip_info

IP information

| Name | Type | Description |
|---|---|---|
| address | string | IPv4 or IPv6 address |
| family | string | IPv4 or IPv6 |
| netmask | string | Input as netmask length (16) or IPv4 mask (255.255.0.0). For IPv6, the default value is 64 with a valid range of 1 to 127. Output is always netmask length. |

ipspace

Either the UUID or name must be supplied on POST for cluster-scoped objects.

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | IPspace name |
| uuid | string | IPspace UUID |

broadcast_domain

Broadcast domain UUID along with a readable name.

| Name | Type | Description |
|---|---|---|
| _links | _links | |

| Name | Type | Description |
|---|---|---|
| name | string | Name of the broadcast domain, scoped to its IPspace |
| uuid | string | Broadcast domain UUID |

home_node

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | |
| uuid | string | |

node

| Name | Type | Description |
|---|---|---|
| name | string | Name of node on which the port is located. |

home_port

Port UUID along with readable names. Either the UUID or both names may be supplied on input.

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | |
| node | node | |
| uuid | string | |

node

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | |
| uuid | string | |

port

Port UUID along with readable names. Either the UUID or both names may be supplied on input.

| Name | Type | Description |
|---|---|---|
| _links | _links | |

| Name | Type | Description |
| --- | --- | --- |
| name | string | |
| node | node | |
| uuid | string | |

location

Current or home location can be modified. Specifying a port implies a node. Specifying a node allows an appropriate port to be automatically selected. Ports are not valid and not shown for VIP interfaces. For POST, broadcast_domain can be specified alone or with home_node. For PATCH, set is_home to true to revert a LIF back to its home port.

| Name | Type | Description |
| --- | --- | --- |
| auto_revert | boolean | |
| broadcast_domain | broadcast_domain | Broadcast domain UUID along with a readable name. |
| failover | string | Defines where an interface may failover. |
| home_node | home_node | |
| home_port | home_port | Port UUID along with readable names. Either the UUID or both names may be supplied on input. |
| is_home | boolean | |
| node | node | |
| port | port | Port UUID along with readable names. Either the UUID or both names may be supplied on input. |

throughput

The rate of throughput bytes per second observed at the interface.

| Name | Type | Description |
| --- | --- | --- |
| read | integer | Performance metric for read I/O operations. |
| total | integer | Performance metric aggregated over all types of I/O operations. |
| write | integer | Peformance metric for write I/O operations. |

metric

The most recent sample of I/O metrics for the interface.

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| duration | string | The duration over which this sample is calculated. The time durations are represented in the ISO-8601 standard format. Samples can be calculated over the following durations: |
| status | string | Errors associated with the sample. For example, if the aggregation of data over multiple nodes fails, then any partial errors might return "ok" on success or "error" on an internal uncategorized failure. Whenever a sample collection is missed but done at a later time, it is back filled to the previous 15 second timestamp and tagged with "backfilled_data". "inconsistent_delta_time" is encountered when the time between two collections is not the same for all nodes. Therefore, the aggregated value might be over or under inflated. "Negative_delta" is returned when an expected monotonically increasing value has decreased in value. "inconsistent_old_data" is returned when one or more nodes do not have the latest data. |
| throughput | throughput | The rate of throughput bytes per second observed at the interface. |
| timestamp | string | The timestamp of the performance data. |

service_policy

| Name | Type | Description |
|---|---|---|
| _links | _links | |

| Name | Type | Description |
| --- | --- | --- |
| name | string | |
| uuid | string | |

throughput_raw

Throughput bytes observed at the interface. This can be used along with delta time to calculate the rate of throughput bytes per unit of time.

| Name | Type | Description |
| --- | --- | --- |
| read | integer | Performance metric for read I/O operations. |
| total | integer | Performance metric aggregated over all types of I/O operations. |
| write | integer | Peformance metric for write I/O operations. |

statistics

The real time I/O statistics for the interface.

| Name | Type | Description |
| --- | --- | --- |
| status | string | Errors associated with the sample. For example, if the aggregation of data over multiple nodes fails, then any partial errors might return "ok" on success or "error" on an internal uncategorized failure. Whenever a sample collection is missed but done at a later time, it is back filled to the previous 15 second timestamp and tagged with "backfilled_data". "inconsistent_delta_time" is encountered when the time between two collections is not the same for all nodes. Therefore, the aggregated value might be over or under inflated. "Negative_delta" is returned when an expected monotonically increasing value has decreased in value. "inconsistent_old_data" is returned when one or more nodes do not have the latest data. |
| throughput_raw | throughput_raw | Throughput bytes observed at the interface. This can be used along with delta time to calculate the rate of throughput bytes per unit of time. |
| timestamp | string | The timestamp of the performance data. |

svm

Applies only to SVM-scoped objects. Either the UUID or name must be supplied on POST.

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| name | string | The name of the SVM. |
| uuid | string | The unique identifier of the SVM. |

error_arguments

| Name | Type | Description |
|---|---|---|
| code | string | Argument code |
| message | string | Message argument |

error

| Name | Type | Description |
|---|---|---|
| arguments | array[error_arguments] | Message arguments |
| code | string | Error code |
| message | string | Error message |
| target | string | The target parameter that caused the error. |

# Update an IP interface

PATCH `/network/ip/interfaces/{uuid}`

**Introduced In:** 9.6

Updates an IP interface.

## Related ONTAP commands

- `network interface migrate`
- `network interface modify`
- `network interface rename`
- `network interface revert`

## Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| uuid | string | path | True | IP interface UUID |

## Request Body

| Name | Type | Description |
|---|---|---|
| _links | _links | |

| Name | Type | Description |
| --- | --- | --- |
| ddns_enabled | boolean | Indicates whether or not dynamic DNS updates are enabled. Defaults to true if the interface supports "data_nfs" or "data_cifs" services, otherwise false. |
| dns_zone | string | Fully qualified DNS zone name |
| enabled | boolean | The administrative state of the interface. |
| ip | ip_info | IP information |
| ipspace | ipspace | Either the UUID or name must be supplied on POST for cluster-scoped objects. |
| location | location | Current or home location can be modified. Specifying a port implies a node. Specifying a node allows an appropriate port to be automatically selected. Ports are not valid and not shown for VIP interfaces. For POST, broadcast_domain can be specified alone or with home_node. For PATCH, set is_home to true to revert a LIF back to its home port. |
| metric | metric | The most recent sample of I/O metrics for the interface. |
| name | string | Interface name |
| probe_port | integer | Probe port for Cloud load balancer |
| rdma_protocols | array[string] | Supported RDMA offload protocols |
| scope | string | Set to "svm" for interfaces owned by an SVM. Otherwise, set to "cluster". |
| service_policy | service_policy | |
| services | array[string] | The services associated with the interface. |

| Name | Type | Description |
| --- | --- | --- |
| state | string | The operational state of the interface. |
| statistics | statistics | The real time I/O statistics for the interface. |
| svm | svm | Applies only to SVM-scoped objects. Either the UUID or name must be supplied on POST. |
| uuid | string | The UUID that uniquely identifies the interface. |
| vip | boolean | True for a VIP interface, whose location is announced via BGP. |

**Example request**


```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "dns_zone": "storage.company.com",
  "ip": {
    "address": "10.10.10.7",
    "family": "ipv4",
    "netmask": "24"
  },
  "ipspace": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "exchange",
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
  },
  "location": {
    "broadcast_domain": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "bd1",
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "failover": "home_port_only",
    "home_node": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "node1",
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "home_port": {
      "_links": {
        "self": {
```

```
          "href": "/api/resourcelink"
        }
      },
      "name": "e1b",
      "node": {
        "name": "node1"
      },
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "node": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "node1",
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "port": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "e1b",
      "node": {
        "name": "node1"
      },
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    }
  },
  "metric": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "duration": "PT15S",
    "status": "ok",
    "throughput": {
      "read": 200,
      "total": 1000,
      "write": 100
    },
    "timestamp": "2017-01-25T11:20:13Z"
  },
```

```
  "name": "dataLif1",
  "probe_port": 64001,
  "rdma_protocols": {
  },
  "scope": "svm",
  "service_policy": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "default-intercluster",
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
  },
  "services": {
  },
  "state": "up",
  "statistics": {
    "status": "ok",
    "throughput_raw": {
      "read": 200,
      "total": 1000,
      "write": 100
    },
    "timestamp": "2017-01-25T11:20:13Z"
  },
  "svm": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "svm1",
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  },
  "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
}
```


## Response

```
Status: 200, Ok
```

## Error

```
Status: Default
```

ONTAP Error Response Codes

| Error Code | Description |
|---|---|
| 1376663 | Cannot add interface to DNS zone because all interfaces from a single DNS zone must be in the same SVM. |
| 1376963 | Duplicate IP address. |
| 1376997 | Interface failed to migrate because the node hosting the port is not healthy. |
| 1376998 | The specified location.node does not own any ports in the same broadcast domain as the home port of the interface. |
| 1376999 | Interface failed to migrate because port is in the down admin state. |
| 1377607 | The specified location.port is not in the same broadcast domain as the home port of the interface. |
| 1966138 | The same IP address may not be used for both a mgmt interface and a gateway address. |
| 1966141 | Invalid DNS zone name. |
| 1966142 | Only data LIFs can be assigned a DNS zone. |
| 1966197 | Migration of cluster interfaces must be done from the local node. |
| 1966267 | IPv6 addresses must have a prefix length between 1 and 127. |
| 1966269 | IPv4 addresses must have a prefix length between 1 and 32. |
| 1966476 | DNS Update is supported only on data interfaces. |
| 1966477 | DNS Update is supported only on interfaces configured with the NFS or CIFS protocol. |
| 1967106 | The specified location.home_port.name does not match the specified port name of location.home_port.uuid. |
| 1967107 | The specified location.home_port.uuid is not valid. |
| 1967111 | A home node must be specified by at least one location.home_node, location.home_port, or location.broadcast_domain field. |
| 1967113 | The specified location.port.name does not match the port name of location.port.uuid. |

| Error Code | Description |
| --- | --- |
| 1967114 | The specified location.port.uuid is not valid. |
| 1967115 | The specified location.node.name does not match the node name of location.node.uuid. |
| 1967116 | The specified location.port.node.name does not match the node name of location.node.uuid. |
| 1967117 | The specified location.port.node.name does not match location.node.name. |
| 1967118 | A node must be specified by at least one location.node or location.port field. |
| 1967119 | The specified location.node.name does not match the node name of location.port.uuid. |
| 1967120 | The specified service_policy.name does not match the specified service policy name of service_policy.uuid. |
| 1967121 | The specified service_policy.uuid is not valid. |
| 1967125 | You cannot patch the "location.node" or "location.port" fields to migrate interfaces using the iSCSI data protocol. Instead perform the following PATCH operations on the interface: set the "enabled" field to "false"; change one or more "location.home_port" fields to migrate the interface; and then set the "enabled" field to "true". |
| 1967129 | The specified location.home_port.uuid is not valid. |
| 1967130 | The specified location.home_port.name is not valid. |
| 1967131 | The specified location.home_port.uuid and location.home_port.name are not valid. |
| 1967132 | The specified location.port.uuid is not valid. |
| 1967133 | The specified location.port.name is not valid. |
| 1967134 | The specified location.port.uuid and location.port.name are not valid. |
| 1967138 | Cannot patch port for a VIP interface. The specified parameter location.port.uuid is not valid. |
| 1967139 | Cannot patch port for a VIP interface. The specified parameter location.port.name is not valid. |
| 1967140 | Cannot patch port for a VIP interface. The specified parameters location.port.uuid and location.port.name are not valid. |
| 1967141 | Cannot patch home_port for a VIP interface. The specified parameter location.home_port.uuid is not valid. |

| Error Code | Description |
| --- | --- |
| 1967142 | Cannot patch home_port for a VIP interface. The specified parameter location.home_port.name is not valid. |
| 1967143 | Cannot patch home_port for a VIP interface. The specified parameters location.home_port.uuid and location.home_port.name are not valid. |
| 1967145 | The specified location.failover is not valid. |
| 1967153 | No suitable port exists on location.home_node to host the interface. |
| 1967380 | Cannot patch home_port for a VIP interface. The specified parameter location.home_port.node.name is not valid. Consider using location.home_node.name instead. |
| 1967386 | Cannot patch port for a VIP interface. The specified parameter location.port.node.name is not valid. Consider using location.node.name instead. |
| 1967387 | The specified IP address is in use by a subnet in this IPspace. |
| 1967389 | Patching location.is_home to the value "false" is not supported. The value "true" would revert a network interface to its home port if the current value is "false". |
| 1967390 | Cannot patch a LIF revert as it requires an effective cluster version of 9.9.1 or later. |
| 1967391 | Patching the DNS zone requires an effective cluster version of 9.9.1 or later. |
| 1967392 | Patching the DDNS enable parameter requires an effective cluster version of 9.9.1 or later. |
| 53281065 | The service_policy does not exist in the SVM. |
| 53281086 | LIF would exceed the maximum number of supported intercluster LIFs in IPspace. |
| 53281089 | LIF on SVM cannot be updated to use service policy because that service policy includes SAN services and the target LIF is not home. |

## Definitions

**See Definitions**

href

| Name | Type | Description |
|---|---|---|
| href | string | |

_links

| Name | Type | Description |
|---|---|---|
| self | href | |

ip_info

IP information

| Name | Type | Description |
|---|---|---|
| address | string | IPv4 or IPv6 address |
| family | string | IPv4 or IPv6 |
| netmask | string | Input as netmask length (16) or IPv4 mask (255.255.0.0). For IPv6, the default value is 64 with a valid range of 1 to 127. Output is always netmask length. |

ipspace

Either the UUID or name must be supplied on POST for cluster-scoped objects.

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | IPspace name |
| uuid | string | IPspace UUID |

broadcast_domain

Broadcast domain UUID along with a readable name.

| Name | Type | Description |
|---|---|---|
| _links | _links | |

| Name | Type | Description |
| --- | --- | --- |
| name | string | Name of the broadcast domain, scoped to its IPspace |
| uuid | string | Broadcast domain UUID |

home_node

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| name | string | |
| uuid | string | |

node

| Name | Type | Description |
| --- | --- | --- |
| name | string | Name of node on which the port is located. |

home_port

Port UUID along with readable names. Either the UUID or both names may be supplied on input.

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| name | string | |
| node | node | |
| uuid | string | |

node

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| name | string | |
| uuid | string | |

port

Port UUID along with readable names. Either the UUID or both names may be supplied on input.

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |

| Name | Type | Description |
|---|---|---|
| name | string | |
| node | node | |
| uuid | string | |

location

Current or home location can be modified. Specifying a port implies a node. Specifying a node allows an appropriate port to be automatically selected. Ports are not valid and not shown for VIP interfaces. For POST, broadcast_domain can be specified alone or with home_node. For PATCH, set is_home to true to revert a LIF back to its home port.

| Name | Type | Description |
|---|---|---|
| auto_revert | boolean | |
| broadcast_domain | broadcast_domain | Broadcast domain UUID along with a readable name. |
| failover | string | Defines where an interface may failover. |
| home_node | home_node | |
| home_port | home_port | Port UUID along with readable names. Either the UUID or both names may be supplied on input. |
| is_home | boolean | |
| node | node | |
| port | port | Port UUID along with readable names. Either the UUID or both names may be supplied on input. |

throughput

The rate of throughput bytes per second observed at the interface.

| Name | Type | Description |
|---|---|---|
| read | integer | Performance metric for read I/O operations. |
| total | integer | Performance metric aggregated over all types of I/O operations. |
| write | integer | Peformance metric for write I/O operations. |

metric

The most recent sample of I/O metrics for the interface.

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| duration | string | The duration over which this sample is calculated. The time durations are represented in the ISO-8601 standard format. Samples can be calculated over the following durations: |
| status | string | Errors associated with the sample. For example, if the aggregation of data over multiple nodes fails, then any partial errors might return "ok" on success or "error" on an internal uncategorized failure. Whenever a sample collection is missed but done at a later time, it is back filled to the previous 15 second timestamp and tagged with "backfilled_data". "inconsistent_delta_time" is encountered when the time between two collections is not the same for all nodes. Therefore, the aggregated value might be over or under inflated. "Negative_delta" is returned when an expected monotonically increasing value has decreased in value. "inconsistent_old_data" is returned when one or more nodes do not have the latest data. |
| throughput | throughput | The rate of throughput bytes per second observed at the interface. |
| timestamp | string | The timestamp of the performance data. |

service_policy

| Name | Type | Description |
|---|---|---|
| _links | _links | |

| Name | Type | Description |
| --- | --- | --- |
| name | string | |
| uuid | string | |

throughput_raw

Throughput bytes observed at the interface. This can be used along with delta time to calculate the rate of throughput bytes per unit of time.

| Name | Type | Description |
| --- | --- | --- |
| read | integer | Performance metric for read I/O operations. |
| total | integer | Performance metric aggregated over all types of I/O operations. |
| write | integer | Peformance metric for write I/O operations. |

statistics

The real time I/O statistics for the interface.

| Name | Type | Description |
| --- | --- | --- |
| status | string | Errors associated with the sample. For example, if the aggregation of data over multiple nodes fails, then any partial errors might return "ok" on success or "error" on an internal uncategorized failure. Whenever a sample collection is missed but done at a later time, it is back filled to the previous 15 second timestamp and tagged with "backfilled_data". "inconsistent_delta_time" is encountered when the time between two collections is not the same for all nodes. Therefore, the aggregated value might be over or under inflated. "Negative_delta" is returned when an expected monotonically increasing value has decreased in value. "inconsistent_old_data" is returned when one or more nodes do not have the latest data. |
| throughput_raw | throughput_raw | Throughput bytes observed at the interface. This can be used along with delta time to calculate the rate of throughput bytes per unit of time. |
| timestamp | string | The timestamp of the performance data. |

svm

Applies only to SVM-scoped objects. Either the UUID or name must be supplied on POST.

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| name | string | The name of the SVM. |
| uuid | string | The unique identifier of the SVM. |

ip_interface

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| ddns_enabled | boolean | Indicates whether or not dynamic DNS updates are enabled. Defaults to true if the interface supports "data_nfs" or "data_cifs" services, otherwise false. |
| dns_zone | string | Fully qualified DNS zone name |
| enabled | boolean | The administrative state of the interface. |
| ip | ip_info | IP information |
| ipspace | ipspace | Either the UUID or name must be supplied on POST for cluster-scoped objects. |
| location | location | Current or home location can be modified. Specifying a port implies a node. Specifying a node allows an appropriate port to be automatically selected. Ports are not valid and not shown for VIP interfaces. For POST, broadcast_domain can be specified alone or with home_node. For PATCH, set is_home to true to revert a LIF back to its home port. |
| metric | metric | The most recent sample of I/O metrics for the interface. |
| name | string | Interface name |
| probe_port | integer | Probe port for Cloud load balancer |
| rdma_protocols | array[string] | Supported RDMA offload protocols |
| scope | string | Set to "svm" for interfaces owned by an SVM. Otherwise, set to "cluster". |
| service_policy | service_policy | |

| Name | Type | Description |
|---|---|---|
| services | array[string] | The services associated with the interface. |
| state | string | The operational state of the interface. |
| statistics | statistics | The real time I/O statistics for the interface. |
| svm | svm | Applies only to SVM-scoped objects. Either the UUID or name must be supplied on POST. |
| uuid | string | The UUID that uniquely identifies the interface. |
| vip | boolean | True for a VIP interface, whose location is announced via BGP. |

# Retrieve interface historical performance metrics

GET `/network/ip/interfaces/{uuid}/metrics`

**Introduced In:** 9.8

Retrieves historical performance metrics for an interface.

## Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| throughput.read | integer | query | False | Filter by throughput.read |
| throughput.total | integer | query | False | Filter by throughput.total |
| throughput.write | integer | query | False | Filter by throughput.write |
| status | string | query | False | Filter by status |
| timestamp | string | query | False | Filter by timestamp |
| duration | string | query | False | Filter by duration |

| Name | Type | In | Required | Description |
| --- | --- | --- | --- | --- |
| uuid | string | path | True | Unique identifier of the interface. |
| interval | string | query | False | The time range for the data. Examples can be 1h, 1d, 1m, 1w, 1y. The period for each time range is as follows:<br>• 1h: Metrics over the most recent hour sampled over 15 seconds.<br>• 1d: Metrics over the most recent day sampled over 5 minutes.<br>• 1w: Metrics over the most recent week sampled over 30 minutes.<br>• 1m: Metrics over the most recent month sampled over 2 hours.<br>• 1y: Metrics over the most recent year sampled over a day.<br>• Default value: 1<br>• enum: ["1h", "1d", "1w", "1m", "1y"] |

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| return_timeout | integer | query | False | The number of seconds to allow the call to execute before returning. When iterating over a collection, the default is 15 seconds. ONTAP returns earlier if either max records or the end of the collection is reached.<br>• Default value: 1<br>• Max value: 120<br>• Min value: 0 |
| fields | array[string] | query | False | Specify the fields to return. |
| max_records | integer | query | False | Limit the number of records returned. |
| order_by | array[string] | query | False | Order results by specified fields and optional [asc |
| desc] direction. Default direction is 'asc' for ascending. | return_records | boolean | query | False |

## Response

```
Status: 200, Ok
```

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| num_records | integer | Number of records |
| records | array[records] | |

#### Example response

```
{
  "_links": {
    "next": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "records": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "duration": "PT15S",
    "status": "ok",
    "throughput": {
      "read": 200,
      "total": 1000,
      "write": 100
    },
    "timestamp": "2017-01-25T11:20:13Z",
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
  }
}
```

## Error

```
Status: Default, Error
```

| Name | Type | Description |
|---|---|---|
| error | error | |

**Example error**

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

## Definitions

**See Definitions**

href

| Name | Type | Description |
|---|---|---|
| href | string | |

_links

| Name | Type | Description |
|---|---|---|
| next | href | |
| self | href | |

_links

| Name | Type | Description |
|---|---|---|
| self | href | |

throughput

The rate of throughput bytes per second observed at the interface.

| Name | Type | Description |
|---|---|---|
| read | integer | Performance metric for read I/O operations. |
| total | integer | Performance metric aggregated over all types of I/O operations. |
| write | integer | Peformance metric for write I/O operations. |

records

Throughput performance for the interfaces.

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| duration | string | The duration over which this sample is calculated. The time durations are represented in the ISO-8601 standard format. Samples can be calculated over the following durations: |

| Name | Type | Description |
| --- | --- | --- |
| status | string | Errors associated with the sample. For example, if the aggregation of data over multiple nodes fails, then any partial errors might return "ok" on success or "error" on an internal uncategorized failure. Whenever a sample collection is missed but done at a later time, it is back filled to the previous 15 second timestamp and tagged with "backfilled_data". "inconsistent_delta_time" is encountered when the time between two collections is not the same for all nodes. Therefore, the aggregated value might be over or under inflated. "Negative_delta" is returned when an expected monotonically increasing value has decreased in value. "inconsistent_old_data" is returned when one or more nodes do not have the latest data. |
| throughput | throughput | The rate of throughput bytes per second observed at the interface. |
| timestamp | string | The timestamp of the performance data. |
| uuid | string | The UUID that uniquely identifies the interface. |

error_arguments

| Name | Type | Description |
| --- | --- | --- |
| code | string | Argument code |
| message | string | Message argument |

error

| Name | Type | Description |
| --- | --- | --- |
| arguments | array[error_arguments] | Message arguments |

| Name | Type | Description |
| --- | --- | --- |
| code | string | Error code |
| message | string | Error message |
| target | string | The target parameter that caused the error. |

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